

Homework 1

PENG Qiheng

Student ID 225040065

September 19, 2026

Problem 1. Conditional probability

(a) By the law of total probability,

$$P(A) = P(A | W)P(W) + P(A | F)P(F) = \frac{1}{5} \cdot \frac{3}{4} + \frac{4}{5} \cdot \frac{1}{4} = \boxed{\frac{7}{20}}.$$

(b) By Bayes' formula,

$$P(F | A) = \frac{P(A | F)P(F)}{P(A)} = \frac{(4/5)(1/4)}{7/20} = \boxed{\frac{4}{7}}.$$

Problem 2. Transition matrices and the Markov property

(a) A transition matrix must have nonnegative entries and each row must sum to one.

Thus A is valid. Matrix B is invalid because its row sums are 1.1 and 0.9. Matrix C is invalid because $c_{12} = -0.1 < 0$ (and $c_{11} = 1.1 > 1$).

(b) By the Markov property and time homogeneity,

$$P(X_2 = 1 | X_1 = 0, X_0 = 1) = P(X_2 = 1 | X_1 = 0) = a_{01} = \boxed{0.4}.$$

(c) No. The next state is conditionally independent of earlier states *given the present state*; it can still depend on the present state. For example, under A ,

$$P(X_{n+1} = 1 | X_n = 0) = 0.4, \quad P(X_{n+1} = 1 | X_n = 1) = 0.8.$$

Whenever both conditioning states have positive probability, these unequal probabilities rule out independence.

Problem 3. **A weather model**

(a) In state order (R, D) ,

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

(b) Matrix multiplication gives

$$P^2 = \begin{pmatrix} (3/4)^2 + (1/4)^2 & 2(3/4)(1/4) \\ 2(3/4)(1/4) & (1/4)^2 + (3/4)^2 \end{pmatrix} = \begin{pmatrix} 5/8 & 3/8 \\ 3/8 & 5/8 \end{pmatrix}.$$

(c) The two-step transition probability is

$$P(X_2 = R \mid X_0 = R) = (P^2)_{RR} = \boxed{\frac{5}{8}}.$$

The specified path has probability

$$P(X_1 = D, X_2 = R \mid X_0 = R) = p_{RD}p_{DR} = \frac{1}{4} \cdot \frac{1}{4} = \boxed{\frac{1}{16}}.$$

Problem 4. **An initial distribution**

Solution. Using row vectors in state order $(0, 1, 2)$,

$$\begin{aligned} \mu_1 &= \mu_0 P = \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4}, \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}, \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} \right) = \boxed{\left(\frac{3}{8}, \frac{1}{2}, \frac{1}{8} \right)}, \\ \mu_2 &= \mu_1 P = \left(\frac{3}{16} + \frac{1}{8}, \frac{3}{16} + \frac{1}{4} + \frac{1}{16}, \frac{1}{8} + \frac{1}{16} \right) = \boxed{\left(\frac{5}{16}, \frac{1}{2}, \frac{3}{16} \right)}. \end{aligned}$$

Consequently,

$$E[X_2] = 0 \cdot \frac{5}{16} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{16} = \boxed{\frac{7}{8}}.$$

Problem 5. An urn with two balls

- (a) The state space is $\{0, 1, 2\}$ and $X_0 = 2$. From state i , selecting a red ball decreases the count with probability $i/2$; selecting a blue ball increases it with probability $(2 - i)/2$. Hence, in order $(0, 1, 2)$,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

- (b) Since $X_1 = 1$ with probability one,

$$\mathcal{L}(X_2) = \left(\frac{1}{2}, 0, \frac{1}{2} \right) \quad \text{in order } (0, 1, 2).$$

- (c) Just before the third selection, the urn is described by X_2 . Conditional on $X_2 = i$, the chance of selecting red is $i/2$. Therefore,

$$P(\text{third selected ball is red}) = \sum_{i=0}^2 \frac{i}{2} P(X_2 = i) = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \boxed{\frac{1}{2}}.$$

Problem 6. Two consecutive heads

- (a) A tail resets state 0 or 1 to 0, whereas a head advances it by one. State 2 is absorbing, so

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (b) Starting from $\mu_0 = (1, 0, 0)$, repeated multiplication gives

$$\mu_2 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right), \quad \mu_3 = \left(\frac{3}{8}, \frac{1}{4}, \frac{3}{8} \right), \quad \mu_4 = \left(\frac{5}{16}, \frac{3}{16}, \frac{1}{2} \right).$$

Since state 2 records success by that time, $\boxed{P(N \leq 4) = \mu_4(2) = 1/2}$.

- (c) Absorption first at time 4 has probability

$$P(N = 4) = P(N \leq 4) - P(N \leq 3) = \frac{1}{2} - \frac{3}{8} = \boxed{\frac{1}{8}}.$$

Equivalently, the only such sequences are $HTHH$ and $TTHH$, each with probability $1/16$.

Problem 7. First-step analysis in a fair game

(a) The boundary values are $\boxed{h_0 = 0, h_3 = 1}$.

(b) Conditioning on the first play gives

$$h_1 = \frac{1}{2}h_0 + \frac{1}{2}h_2 = \frac{1}{2}h_2, \quad h_2 = \frac{1}{2}h_1 + \frac{1}{2}h_3 = \frac{1}{2}h_1 + \frac{1}{2}.$$

(c) Substituting the second equation into the first,

$$h_1 = \frac{1}{4}h_1 + \frac{1}{4}, \quad \boxed{h_1 = \frac{1}{3}, \quad h_2 = \frac{2}{3}}.$$

Absorption at a boundary occurs almost surely. Thus, starting with one unit, the ruin probability is $\boxed{1 - h_1 = 2/3}$.

Problem 8. A biased game

(a) For upper boundary M , the gambler's ruin formula is

$$h_i = \frac{1 - (q/p)^i}{1 - (q/p)^M} \quad (p \neq q).$$

Here $p = 2/3$, $q = 1/3$, $M = 4$, and $i = 1$. Therefore,

$$P(\text{reach 4 before 0}) = \frac{1 - 1/2}{1 - (1/2)^4} = \boxed{\frac{8}{15}},$$

$$P(\text{ruin}) = 1 - \frac{8}{15} = \boxed{\frac{7}{15}}.$$

(b) In the fair case, $h_i = i/M$, so the success probability is $\boxed{1/4}$. The upward bias increases it from $1/4$ to $8/15$, a difference of

$$\frac{8}{15} - \frac{1}{4} = \frac{17}{60} > 0.$$

Problem 9. **Communicating classes**

- (a) The communicating classes are

$$\boxed{\{0, 1\}, \quad \{2\}, \quad \{3\}}.$$

Classes $\{0, 1\}$ and $\{3\}$ are closed. Class $\{2\}$ is not closed because state 2 can move to 0 or 3.

- (b) States 0 and 1 are recurrent (indeed, positive recurrent) because they form a finite closed irreducible class. State 3 is absorbing and hence also positive recurrent. State 2 is transient: it leaves for 0 or 3 with probability $1/2$ on the first step, after which it can never return. Its positive-time return probability is only $p_{22} = 1/2 < 1$.

- (c) Yes. The path $2 \rightarrow 0 \rightarrow 1$ has probability

$$p_{20}p_{01} = \frac{1}{4} \cdot \frac{1}{2} = \boxed{\frac{1}{8}} > 0.$$

Problem 10. **First return versus a visit at a fixed time**

- (a) Staying in state 2 for one step counts as a positive return:

$$\boxed{P(T_2^+ = 1 \mid X_0 = 2) = p_{22} = \frac{1}{2}}.$$

If the first step leaves 2, the chain enters one of the closed classes and can never return. Hence

$$\boxed{P(T_2^+ < \infty \mid X_0 = 2) = \frac{1}{2}}.$$

- (b) A first return at time 2 would require leaving 2 at time 1, which prevents any return. Therefore,

$$\boxed{P(T_2^+ = 2 \mid X_0 = 2) = 0}.$$

A visit at time 2 is possible via $2 \rightarrow 2 \rightarrow 2$, giving

$$\boxed{P(X_2 = 2 \mid X_0 = 2) = p_{22}^2 = \frac{1}{4}}.$$

On this path the first return is at time 1. A fixed-time visit permits earlier returns; a first return does not.

Problem 11. **An unrestricted random walk**

- (a) Returning at time 2 requires increments $(+1, -1)$ or $(-1, +1)$. Thus

$$P(X_2 = 0 \mid X_0 = 0) = pq + qp = 2 \cdot \frac{3}{4} \cdot \frac{1}{4} = \boxed{\frac{3}{8}}.$$

- (b) Using the stated return-probability formula,

$$f_0 = 2 \min\{p, q\} = 2 \cdot \frac{1}{4} = \boxed{\frac{1}{2}}.$$

Since $f_0 < 1$, state 0 is **transient**.

- (c) When $p = q = 1/2$, state 0 is **null recurrent**. Its positive return time $T_0^+ = \min\{n \geq 1 : X_n = 0\}$ is finite almost surely, but $\boxed{E[T_0^+ \mid X_0 = 0] = \infty}$.

Problem 12. **A stationary distribution**

- (a) The balance and normalization equations give

$$\pi_W = \frac{3}{4}\pi_W + \frac{1}{2}\pi_F, \quad \pi_W + \pi_F = 1.$$

Hence $\pi_W = 2\pi_F$, and in state order (W, F) ,

$$\boxed{\pi = \left(\frac{2}{3}, \frac{1}{3}\right)}.$$

- (b) The chain is finite and irreducible, so the long-run proportion of faulty days is $\boxed{\pi_F = 1/3}$, regardless of its initial state.

- (c) Let $T_W^+ = \min\{n \geq 1 : X_n = W\}$. The mean return-time formula for a positive recurrent state gives

$$\boxed{E[T_W^+ \mid X_0 = W] = \frac{1}{\pi_W} = \frac{3}{2} \text{ days}}.$$

To check directly, let m_F be the expected time to hit W from F . Then $m_F = 1 + \frac{1}{2}m_F$, so $m_F = 2$ and $E[T_W^+ \mid X_0 = W] = 1 + \frac{1}{4}m_F = 3/2$.

Problem 13. **A three-state chain**

- (a) There are positive-probability transitions in both directions along $0 \leftrightarrow 1 \leftrightarrow 2$, so all states communicate and the chain is irreducible. Each state has $p_{ii} = 1/2 > 0$, so a return in one step is possible and every state has period 1. Thus the chain is aperiodic.
- (b) Writing $\pi = (\pi_0, \pi_1, \pi_2)$, the equations are

$$\begin{cases} \pi_0 = \frac{1}{2}\pi_0 + \frac{1}{4}\pi_1, \\ \pi_1 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{1}{2}\pi_2, \\ \pi_2 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2, \\ \pi_0 + \pi_1 + \pi_2 = 1. \end{cases}$$

The first and third equations yield $\pi_0 = \pi_2 = \pi_1/2$. Normalization gives $2\pi_1 = 1$, so $\pi = (1/4, 1/2, 1/4)$; this also satisfies the second equation.

- (c) A finite irreducible chain is positive recurrent. Together with aperiodicity, this ensures convergence to its unique stationary distribution. Therefore,

$$\lim_{n \rightarrow \infty} P(X_n = 2 \mid X_0 = 0) = \pi_2 = \frac{1}{4}.$$

Problem 14. **Stationarity and periodicity**

- (a) Both states return exactly at positive even times. Their period is $\gcd\{2, 4, 6, \dots\} = 2$. Since $\pi P = (\pi_1, \pi_0)$, stationarity and normalization give $\pi = (1/2, 1/2)$.
- (b) Starting from 0, the chain alternates deterministically:

$$P(X_n = 0 \mid X_0 = 0) = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

This probability has no limit as $n \rightarrow \infty$.

- (c) If $X_0 \sim \pi$, then $\pi P^n = \pi$ for every $n \geq 0$. Hence $P(X_n = 0) = 1/2$ for every $n \geq 0$.

Problem 15. **Long-run average cost**

- (a) Using $\pi = (2/3, 1/3)$ from Problem 12, the long-run average cost per day is

$$\bar{c} = 20\pi_W + 80\pi_F = 20 \cdot \frac{2}{3} + 80 \cdot \frac{1}{3} = \boxed{40 \text{ units}}.$$

- (b) Given that the machine is working today, tomorrow's expected cost is

$$E[c(X_{n+1}) \mid X_n = W] = 20 \cdot \frac{3}{4} + 80 \cdot \frac{1}{4} = \boxed{35 \text{ units}}.$$

- (c) The transition matrix and thus the stationary distribution are unchanged. The increase in long-run average daily cost is

$$(110 - 80)\pi_F = 30 \cdot \frac{1}{3} = \boxed{10 \text{ units}}.$$

The new long-run average is 50 units per day.

Problem 16. **Short conceptual questions**

- (a) **True.** A finite irreducible Markov chain has a unique stationary distribution with $\pi_i > 0$ for every state. Each state is positive recurrent, with mean positive return time $E_i[T_i^+] = 1/\pi_i < \infty$.
- (b) **False.** In Problem 14, start with $\pi = (1/2, 1/2)$. Every marginal distribution remains π , but

$$P(X_0 = 0, X_1 = 0) = 0 \neq P(X_0 = 0)P(X_1 = 0) = \frac{1}{4}.$$

Thus stationarity does not imply independence at consecutive times.

- (c) **False.** The symmetric random walk on $\mathbb{Z}$ is recurrent, but the mean positive return time to 0 is infinite. This is null recurrence; finite mean return time requires positive recurrence.